

1. Let k be selected at random from the set $\{1, 2, 3, 4, 6, 8, 12, 18\}$. If $3x = k$, find the probability that the solution for x is a whole number. Express your answer as a common fraction reduced to lowest terms.
2. Let x be a positive integer such that $x < 100$. For how many distinct values of x is $\sin(2x) + 1 > 2$?
3. If the sides of a triangle have lengths of $3a$, $2b$, and c , with $\angle A$ opposite side of length $3a$, then $pa^2 = qb^2 + rc^2 - cwb \cos(A)$. If p , q , r , and w are all positive integers, find the smallest possible value of $(p + q + r + w)$.
4. If $f(x) = 3x - 4$, then an equation for the inverse can be written as $f^{-1}(x) = \frac{x + k}{w}$. Find the value of $(k - w)$.
5. If x is a positive integer, find the smallest possible value of x such that $\cot(x^\circ)$ and $\cot(-x^\circ)$ are equal numbers.
6. From physics, the height s , in feet, of an object after t seconds that has been thrown straight up from a point s_o feet above ground level with an initial velocity of v_o is $s = -16t^2 + v_o t + s_o$. If a baseball is thrown straight up from 3 feet above ground level with initial velocity of 97 feet per second, find the number of feet in the maximum height that the ball will reach. Express your answer as a **decimal** rounded to the nearest hundredth of a foot.
7. A right circular cylinder has a volume of 12π . The lengths of the radius (r) of the base and the height (h) of the cylinder are both positive integers. Find the sum of all possible distinct values of $(r + h)$ where r and h represent the radius and height of one cylinder.
8. The probability that Mary will have one of the top ten Precalculus scores is 0.4263. The probability that Peter will have one of the top ten Precalculus scores is 0.3642. The probability that both will have scores among the top ten Precalculus scores is 0.1968. Expressed as a **decimal** rounded to 4 significant digits, find the probability that both had scores in the top ten Precalculus scores if it is known that at least one of the two did.
9. Rounded to the nearest year, find the number of years needed for a sum of money to quadruple if invested at 7% annual percentage rate compounded continuously.
10. In $\triangle ABC$, $AB = 7$, and $AC = 8$. $\cos(\angle BAC)$ is a rational number that can be expressed as $\frac{k}{w}$ where k and w are relatively prime positive integers. It is known that $(k + w)$ is the square of a positive integer and that BC is a positive integer. Find the sum of all possible distinct values of BC .

11. A box that contains exactly 35 yellow marbles, 7 green marbles, and 8 red marbles. From this box, 5 marbles are drawn (without replacement) at random. Find the probability that the 5 marbles drawn were exactly 3 yellow, 1 green, and 1 red. Express your answer as a common fraction reduced to lowest terms.
12. **(Multiple Choice)** For your answer write the **capital letter** which corresponds to the correct choice. Given: A parabola whose points are all equidistant from the point $(2, 5)$ and the line whose equation is $x = 7$ and that has a latus rectum of length $16\sqrt{5}$.
- A) The conditions above determine a conic.
B) The conditions above overdetermine a conic (that is, there is no conic meeting all the given conditions).
C) The conditions above underdetermine a conic (that is, there is more than one conic that meets all the given conditions).

Note: Be certain to write the correct capital letter as your answer.

13. On the circle whose equation is $(x - 25)^2 + (y - 30)^2 = 145$ a particle is at $A(26, 42)$ and another particle is at $B(24, 18)$. At the same instant, both particles begin to move on the circle. The particle at A moves at a constant speed counter-clockwise on the circle, and the particle at B moves at a constant speed clockwise on the circle. The two particles meet for the first time at $(17, 39)$. The particle that started at A is moving k times the rate of the particle that started at B . Find the value of k . Express your answer as a **decimal** rounded to the nearest hundredth.
14. In this problem, assume that the standard deviation is calculated according to the standard method of calculating the standard deviation for a set of sample proportions. Also, assume the following table of z -scores with the accompanying standard normal probabilities is accurate. On the table $-2(.0228)$ means that there is a normal probability of .0228 of obtaining a z -score of -2 or less.

$-2(.0228)$	$-1.9(.0287)$	$-1.8(.0359)$	$-1.7(.0446)$	$-1.6(.0548)$
$-1.5(.0668)$	$-1.4(.0735)$	$-1.4(.0808)$	$-1.35(.0885)$	$-1.3(.0968)$
$-1.25(.1056)$	$-1.2(.1151)$	$-1.1(.1357)$	$-1(.1587)$	$-0.9(.1841)$
$-0.8(.2119)$	$-0.75(.2266)$	$-0.7(.2420)$	$-0.6(.2743)$	$-0.5(.3085)$
$-0.4(.3446)$	$-0.3(.3821)$	$-0.25(.4013)$	$-0.1(.4602)$	$0(.5000)$
$0.1(.5398)$	$0.2(.5793)$	$0.25(.5987)$	$0.3(.6179)$	$0.4(.6554)$
$0.5(.6915)$	$0.6(.7257)$	$0.7(.7580)$	$0.75(.7734)$	$0.8(.7881)$
$0.9(.8159)$	$1.0(.8413)$	$1.1(.8643)$	$1.2(.8849)$	$1.25(.8944)$
$1.3(.9032)$	$1.4(.9192)$	$1.5(.9332)$	$1.6(.9452)$	$1.7(.9554)$
$1.75(.9599)$	$1.8(.9641)$	$1.9(.9713)$	$2.0(.9772)$	$2.1(.9821)$
$2.2(.9861)$	$2.25(.9878)$	$2.3(.9893)$	$2.4(.9918)$	$2.5(.9938)$

Assume that 80% of all mathletes know the value of $\sqrt{3}$ to 4 significant digits. In a sample observation, Carol selects 100 mathletes at random. Find the probability that the proportion of those mathletes who know the value of $\sqrt{3}$ to 4 significant digits is between 76% and 82%. Express your answer as a **decimal** rounded to 4 decimal places.

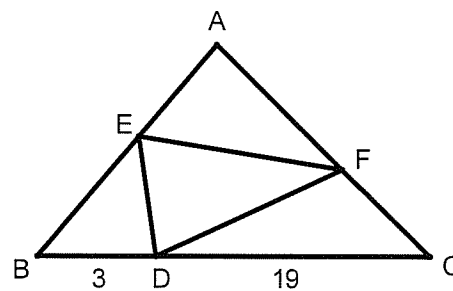
15. Let a , b , and c be positive integers. Find all **ordered triples** (a, b, c) such that

$$\begin{bmatrix} 3 & 4 & 7 \\ 2 & 1 & 8 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 148 \\ 157 \end{bmatrix}. \text{ Be certain to express your answer as } \mathbf{ordered\ triples} \text{ of the form } (a, b, c).$$

16. If $x^4 - 15x^2 + 36$ is factored over the reals into four linear factors, one of those linear factors, in simplest form, is $x - k\sqrt{w}$ where k and w are positive integers greater than one. Find the value of $(5k + 3w)$.
17. If a stick is broken at random into 3 pieces, find the probability that the longest of the 3 pieces is at least three times the shortest of the 3 pieces and that the 3 pieces can serve as the 3 sides of a triangle. Express your answer as a common fraction reduced to lowest terms.
18. If $f(x) = \begin{cases} x - 2 & \text{if } x \leq 0 \\ x^2 + 1 & \text{if } x > 0 \end{cases}$, then the range of f can be expressed in interval notation as $(-\infty, k] \cup (w, \infty)$. Find the value of $(2k + 11w)$.

19. The coordinate axes of the graph of the equation $7x^2 + 8\sqrt{6}xy - 5y^2 - 28 = 0$ are rotated through a positive acute angle (θ) so as to eliminate the xy term. Find the value of $\sin(\theta)$. Express your answer as a **decimal** rounded to the nearest ten-thousandth.

20. In the figure shown, $AB = AC = 15$, $BD = 3$, and $DC = 19$. Points A , E , and B are collinear; points B , D , and C are collinear; and points A , F , and C are collinear. $AE = DE$, and $AF = DF$. Find DE . Express your answer as a **decimal** rounded to the nearest hundredth.



2010 SAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. $\frac{1}{2}$ (Must be this reduced common fraction.)

2. 0 OR "Zero" OR "None"

3. 18

4. 1

5. 90 (Degrees optional.)

6. 150.02 (Must be this decimal.)

7. 18

8. 0.3315 OR .3315 (Must be this decimal.)

9. 20 (Years optional.)

10. 14

11. $\frac{187}{1081}$ (Must be this reduced common fraction.)

12. B (Must be this capital letter.)

13. 0.35 OR .35

14. 0.5328 OR .5328

15. (1,3,19), (6,1,18)

16. 19

17. $\frac{1}{14}$ (Must be this reduced common fraction.)

18. 7

19. 0.4888 OR .4888 (Must be this decimal.)

20. No solution

ITEM ANALYSIS

177 Papers

% correct

3AA Pre-Calcul

1. 91% 11. 2%

2. 82% 12. 5%

3. 33% 13. 10%

4. 85% 14. 14%

5. 53% 15. 14%

6. 69% 16. 63%

7. 71% 17. 1%

8. 5% 18. 7%

9. 66% 19. 3%

10. 14% 20. 0%

159 Papers

(Must be this

decimal.) % correct

4AA Pre-Calcul

1. 93% 11. 38%

2. 89% 12. 72%

3. 56% 13. 17%

4. 92% 14. 15%

5. 67% 15. 29%

6. 83% 16. 78%

(Must be these

ordered triples,

in either order.)

7. 73% 17. 0%

8. 15% 18. 8%

9. 75% 19. 2%

10. 23% 20. 0%

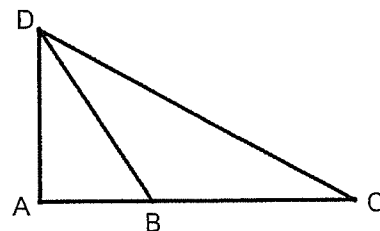
1. Let $i = \sqrt{-1}$. If k and w are real numbers such that $\frac{k + wi}{10 - 24i} = 5$, find the value of $(k + w)$.

2. Let k be a real number. Find the value of $\sin^2(3567894k) + \cos^2(3567894k)$.

3. Find the value of the indicated sum: $\sum_{k=1}^4 (2k + 3)$.

4. Find the sum of the terms of an infinite geometric sequence whose first four terms are 2, -1 , $\frac{1}{2}$, $-\frac{1}{4}$. Express your answer as an improper fraction reduced to lowest terms.

5. In the diagram, points A , B , and C are collinear.
 $\angle DBA = 17.13^\circ$ and $\angle DCA = 11.87^\circ$. If $DA = 967.3$, find BC .
Round your answer to the nearest **integer** and express your answer as that **integer**.

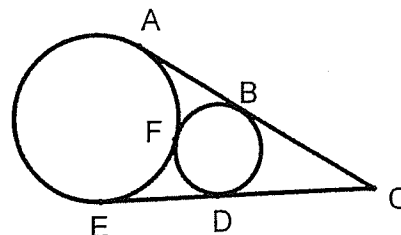


6. Let $\vec{a}$, $\vec{b}$, $\vec{c}$, and $\vec{d}$ represent vectors such that $\vec{a} = (3, 2)$, $\vec{b} = (-8, 13)$, and $\vec{c} = (19, -86)$. Find the **ordered pair** representing $\vec{d}$ if $\vec{a} - \vec{b} = \vec{c} + \vec{d}$.

7. When the sum of the first k terms of the series $1^2 + 2^2 + 3^2 + \cdots + n^2 + \cdots$ is subtracted from the sum of the first k terms of the series $1(2) + 2(3) + 3(4) + \cdots + n(n+1) + \cdots$, the result is 528. Find the value of k .

8. For the equation $x^2 + 11x + k = 0$, the square of the difference between the roots for x is 120 more than the sum of the squares of these roots for x . Find the value of k for which this is true.

9. In the diagram, $\overline{AC}$ and $\overline{EC}$ are common external tangents of the two circles with points of tangency at A , B , D , and E . The circles are tangent at F and have radii of lengths 3 and 8. Then $\sin \angle BCD = \frac{k\sqrt{w}}{f}$ where k , w , and f , are positive integers. Find the smallest possible value of $(k + w + f)$.



10. Let x represent the degree measure of an angle such that $\cot(x) = \sqrt{3}$. If $180^\circ < x < 270^\circ$, find the value of x .
11. Urn A contains five marbles—three of which are orange and two of which are blue. Urn B contains four marbles—three of which are orange and one of which is blue. One of the urns is selected at random, and a marble then is selected at random from that urn. If the marble selected was orange, find the probability that the marble came from Urn B. Express your answer as a common fraction reduced to lowest terms.
12. A parabola has its line of symmetry parallel to the x -axis and has its vertex at $(7, 2)$. The point $(3, -8)$ lies on the parabola. Find the **x -coordinate only** of the focus of this parabola. Express your answer as a **decimal**.
13. Let $\overline{AB}$, $\overline{CD}$, and $\overline{EF}$ be three parallel chords that are non-diameters of a circle on the same side of the center. Let $\overline{GJ}$ be tangent to the circle at H such that $\overline{GJ} \parallel \overline{AB}$. The distance between $\overline{AB}$ and $\overline{CD}$ is equal to the distance between $\overline{CD}$ and $\overline{EF}$ and is also equal to the distance between $\overline{EF}$ and $\overline{GJ}$. If $AB = 24$, then CD **must** be greater than k . Find the largest possible simplified exact value of k .

14. A geometric sequence consists of 10 terms. The sum of the 10 terms is 2343.74976, and the sum of the reciprocals of each of the 10 terms is 1302.0832. Find the product of the 10 terms of the original sequence. Express your answer as an exact decimal.
15. Let k represent a positive integral degree measure such that $\frac{\sin(57^\circ)}{\cos(k^\circ)} = 1$. If $0^\circ < k < 90^\circ$, find the value of k .
16. The sum of the last two terms of an eight term geometric progression of real terms is $\frac{2}{9}$. The sum of the third and fourth terms of this geometric progression is 18. Find the sum of all eight terms of this geometric progression. Express your answer as an improper fraction reduced to lowest terms.
17. In rhombus $ABCD$, $\angle DAB = 52^\circ$. A circle passes through vertices A , B , and D , and intersects diagonal $\overline{AC}$ at E . $CE = 12$. Find the length of the arc of the circle from A to D to B . Express your answer as a decimal rounded to the nearest hundredth.
18. Let $C(n, k) = \frac{n!}{k!(n-k)!}$ where n and k represent positive integers. Find the value of n such that $C(n, 5) = 792$.
19. When $(2x + 3y)^5$ is expanded and completely simplified, the coefficient of one of the terms is 1080. Find the **exponent** of x for that term.
20. By substituting 1, 2, 3, 4, 5, and 6 for x into a polynomial expression with integral coefficients in x , the values are respectively -3 , 10 , 49 , 120 , 229 , and 382 . If $P(x)$ is the polynomial expression with integer coefficients of lowest degree satisfying the given conditions, find $P(43)$.

2011 SAA

Name _____ ANSWERS _____

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

ITEM ANALYSIS
3AA Pre-Calculus
151 Papers
% correct

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 79% 11. 23%
2. 95% 12. 18%
3. 96% 13. 1%
4. 79% 14. 7%

1. _____ -70

11. _____ $\frac{5}{9}$ (Must be this reduced common fraction.)

5. 51% 15. 91%

2. _____ 1

12. _____ 0.75 OR $.75$ (Must be this decimal.)

6. 79% 16. 37%

3. _____ 32

13. _____ $8\sqrt{6}$ (Must be this exact simplified radical.)

7. 69% 17. 1%

4. _____ $\frac{4}{3}$ (Must be this reduced improper fraction.)

14. _____ 18.89568 (Must be this exact decimal.)

9. 1% 19. 74%

5. _____ 1464 (Must be this integer.)

15. _____ 33 (Degrees optional.)

6. _____ $(-8, 75)$ (Must be this ordered pair.)

16. _____ $\frac{1640}{9}$ (Must be this reduced improper fraction.)

7. _____ 32

17. _____ 39.46 (Must be this exact decimal.)
4AA Pre-Calculus

175 Papers
% correct

8. _____ -60

18. _____ 12

1. 79% 11. 39%
2. 98% 12. 26%

9. _____ 167

19. _____ 2

3. 98% 13. 1%
4. 87% 14. 14%
5. 51% 15. 97%
6. 79% 16. 45%

10. _____ 210 (Degrees optional.)

20. _____ 91809

7. 76% 17. 5%
8. 46% 18. 88%
9. 1% 19. 82%
10. 94% 20. 33%

1. Given the sequence $1, 4, 7, 10, \dots, (3n-2)$. If one of the first ten numbers in this sequence is selected at random, find the probability that the number selected is odd. Express your answer as a **common fraction** reduced to lowest terms.

2. **(Multiple Choice)** For your answer write the capital letter that corresponds to the best answer.

For all real x , $\sin(-810+x)^\circ =$

A) $\sin(x)^\circ$ B) $\cos(x)^\circ$ C) $-\sin(x)^\circ$ D) $-\cos(x)^\circ$ E) $\sin(90+x)^\circ$

Note: Be certain to write the correct capital letter as your answer.

3. Find the sum of all distinct values of x such that $|x-7| = |2x+1|$.

4. **(Multiple Choice)** For your answer write the capital letter that corresponds to the best answer.

Let a and b be real numbers, let $i = \sqrt{-1}$, and let $\overline{a+bi}$ be the complex conjugate of $a+bi$. If $(a+bi)(\overline{a+bi}) = 33$ is graphed in a complex plane, then the graph will be:

A) A parabola. B) A circle. C) An ellipse that is **not** a circle.
D) A straight line. E) A hyperbola.

Note: Be certain to write the correct capital letter as your answer.

5. Let $f = \{((4,3),6), ((7,8),9)\}$. Find the sum of all distinct members of the range of f .

6. Find the number of years it will take for a sum of money to double if invested at an annual percentage rate of 8.1% and compounded continuously. Express your answer as a **decimal** rounded to the nearest hundredth of a year.

7. If $\begin{bmatrix} 3 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 7 & 55 \\ 12 & 22 \end{bmatrix}$, find the value of $(a+2b+3c+4d)$.
8. A person writes down 5 different integers at random from the 25 integers from 1 to 25 inclusive. Each of the 25 integers is then called off one at a time in a random order. As soon as all 5 of the person's numbers have been called off, the person yells: "Bingo." Exactly how many of the 25 numbers will have been called off when the probability that the person will yell "Bingo" for the first time is the greatest?
9. Find the exact sum of $\sum_{w=1}^9 (-10)^{(w-3)}$. Express your answer as a **decimal**.
10. An apartment rental company has 3000 apartments available, and 1900 are presently rented at \$950 per month. The company has decided it will only raise or lower the rent per month on all apartments by integral increments of \$40. A survey has shown that, so long as there are apartments available, for each \$40 drop in rent per month for **every** apartment, there will be 99 new tenants. Find the number of dollars in the monthly rent that will maximize total income.
11. In a room with exactly 6 distinct persons, exactly 12 handshakes between 2 distinct persons are made. No two of these handshakes took place between the same 2 persons. Of the persons in the room, exactly two persons shook hands with exactly three other persons. Three distinct persons from these 6 persons are selected at random. Find the probability that those three distinct persons had all shaken hands with each other from among the 12 handshakes. Express your answer as a **common fraction** reduced to lowest terms.
12. Find the number of distinct negative values of x for which $x^9 + 2x^7 = x^4 - 7x^3 - 6x^2 - 7x + 9$.
13. The roots for x of the equation $x^5 - \frac{121}{162}x^4 + bx^3 + cx^2 + dx + e = 0$ form a geometric progression. The sum of the reciprocals of the roots is 242. Find the **absolute value** of e . Express your answer as a **common fraction** reduced to lowest terms.

14. An arithmetic progression has a first term of 3, and its last term is 33. The total population standard deviation (σ_x) of the terms of this arithmetic progression is $\sqrt{105}$. Find the sum of the terms of this arithmetic progression.
15. On April 1, April puts 1 cent in her piggy bank which was empty. On April 2, April puts 2 additional cents in her piggy bank. On April 3, April puts 4 additional cents in her piggy bank. On April 4, April puts 8 additional cents in her piggy bank. She continues this process of doubling the number of cents from the previous day. How many **cents** will April have in her piggy bank after her deposit on the last day of this April month?
16. In *cis* form, the roots of $1 - x^5 = 0$ can be presented as $\text{cis}(k)^\circ$ where $k < 360$. Find the largest possible value of k . (NOTE: $r\text{cis}\theta^\circ = r(\cos\theta^\circ + i\sin\theta^\circ)$)
17. If a stick is broken at random into 3 pieces, find the probability that the longest of the 3 pieces is at least three times the shortest of the 3 pieces. Express your answer as a **common fraction** reduced to lowest terms.
18. Find the radian period of the graph of $y = 3\sin(2x)$.
19. The lengths of the sides of a triangle with an area greater than 4 are respectively 13, 15, and x . If x is a positive integer, find the sum of all distinct possible values of x such that the cosine of the angle opposite the side of length x is a positive common fraction with a denominator less than 37 when reduced to lowest terms.
20. Observe that $1^3 + (-1)^3 + 0^3 = 0^3$
 $2^3 + 1^3 + (-1)^3 = 2^3$
 $9^3 + 15^3 + 12^3 = 18^3$
 $28^3 + 53^3 + 75^3 = 84^3$
 $65^3 + 127^3 + 248^3 = 260^3$
 $126^3 + 249^3 + 615^3 = 630^3$
Should this pattern continue, then the next term would be:
 $k^3 + w^3 + p^3 = 1302^3$
Find the value of $(k + w + p)$.

2012 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. $\frac{1}{2}$ (Must be this reduced common fraction.)

2. D (Must be this capital letter.)

3. -6

4. B (Must be this capital letter.)

5. 15

6. 8.56 (Must be this decimal.)

7. 41

8. 25

9. 909090.91 (Must be this exact decimal.)

10. 870 (\$ optional.)

11. $\frac{1}{2}$ (Must be this reduced common fraction.)

12. 0 OR zero OR none

13. $\frac{1}{1889568}$ (Must be this reduced common fraction.)

14. 108
ITEM ANALYSIS
3AA PRE-CALCULUS
132 Papers
% correct
1. 95% 11. 8%
(Cents optional.)

15. 1073741823
2. 82% 12. 75%
3. 82% 13. 5%
4. 38% 14. 19%
5. 34% 15. 58%

16. 288
6. 66% 16. 12%
17. $\frac{27}{35}$ (Must be this reduced common fraction.)
7. 61% 17. 2%

18. π OR 1π (Must be this exact answer.)
8. 6% 18. 94%

19. 71
9. 76% 19. 10%
10. 57% 20. 15%
4AA Pre-CALCULUS
169 Papers
% correct

20. 1932
1. 98% 11. 13%
2. 82% 12. 76%
3. 89% 13. 18%
4. 52% 14. 33%

5. 43% 15. 68%
6. 76% 16. 27%
7. 65% 17. 1%
8. 10% 18. 92%
9. 85% 19. 21%

1. Find the fifth term of an arithmetic sequence whose first and second terms are respectively 1 and 12.
2. If $f(x) = x^2 + 5$ and $g(x) = (x + 5)^2$, find the value of $(f(3) + g(4))$.
3. Let $i = \sqrt{-1}$. Find the value of $|7 + 4i|$.
4. Let $\|(k, w)\|$ represent the **norm** of the vector represented by (k, w) . If $\|(k, 9)\| = 9\sqrt{5}$ and $k < 0$, find the value of k .
5. The degree measures of the three angles of $\triangle ABC$ are in the ratio of $2:2:5$. The degree measures of the three angles of $\triangle DEF$ are in the ratio of $5:11:14$. One of the six angles from these triangles is selected at random. Find the probability that the angle selected has a degree measure of 40. Express your answer as a common fraction reduced to lowest terms.
6. The first term of a geometric sequence of real terms is 2, and the fifth term of this sequence is 18. If the fourth term of this sequence is negative, find the value of the fourth term of this geometric sequence.
7. Consider the sequence $\{x_n\}$ defined as $x_{(n+1)} = \frac{6x_n}{7} + \frac{5491}{7x_n}$ for integral n such that $n \geq 1$ and for which $x_1 = 11$. As n increases without bound, the limiting value of x_n is k . Find the value of k .

8. The polynomial equation $x^3 + kx^2 + wx + p = 0$ with k , w , and p representing integers has 5 and $1+i$ as two of its roots for x . If $i = \sqrt{-1}$, find the value of p .
9. A plane contains points with coordinates $A(2, -1, 1)$, $B(0, -3, 0)$, and $C(-1, -2, 1)$. A fourth point not on this plane has coordinates $D(5, 2, 4)$. Find the volume of a tetrahedron with vertices at A , B , C , and D .
10. If $\frac{(x+k)!}{x!} - x = 1$ for all possible values of x where x and k represent positive integers, find the value of k .
11. Find the sum of the terms of an infinite geometric sequence whose first four terms are $1, -\frac{1}{4}, \frac{1}{16}, -\frac{1}{64}, \dots$. Express your answer as a common fraction reduced to lowest terms.
12. In a 3-dimensional rectangular coordinate system, we have the points $A(1, 4, 3)$, $B(2, 6, 8)$, and $C(k, w, f)$. If $\overline{AB} \perp \overline{AC}$, find the value of $(k + 2w + 5f)$.
13. Team A plays Team B six times in a season. For each game, the probability that Team A will win is $\frac{3}{5}$. Find the probability that Team A will win at least 3 consecutive games of the six games in the season. Express your answer as a common fraction reduced to lowest terms.
14. Assume that a , b , c , and d are positive, $a + b + c + d = 12$, $a^2 + b^2 + c^2 + d^2 = 50$, and that $c = 2d$. Then the greatest possible value of d can be expressed in the form $\frac{k + w\sqrt{p}}{f}$ where k , w , p , and f are positive integers and $\frac{k + w\sqrt{p}}{f}$ is in simplest radical form. Find the value of $(k + w + p + f)$.

15. The tangent of one of the acute angles of a right triangle is 1. If one of the legs of this right triangle has a length of 16, find the length of the other leg of this right triangle.

16. Sally wishes to produce an 8 digit code (sequence of digits) using only the digits 1, 2, 3, and 4 for each code. How many distinct codes are possible if the number of times the digit 1 is used equals the number of times the digit 2 is used for each code?

17. In $\triangle ABC$, $\sin(\angle BAC) + \cos(\angle BAC) = \frac{89}{65}$. If $\tan(\angle BAC) > 1$, find $\tan(\angle BAC)$. Express your answer as an improper fraction reduced to lowest terms.

18. (Multiple **Choice**) For your answer write the **capital letter** which corresponds to the correct choice.

The graph of $xy = 324$ is symmetric with respect to the:

- A) Origin.
- B) y-axis.
- C) x-axis.
- D) Both the origin and the x-axis.
- E) Both the origin and the y-axis.

Note: Be certain to write the correct capital letter as your answer.

19. Find the value of $\lim_{x \rightarrow \infty} \left(\frac{20x+14}{2x+3} \right)$.

20. Given that $x^2 + x + 1 = 0$ and $P(m) = x^m + \frac{1}{x^m}$. Find the exact value of $P(2013) + P(2014) + P(2015) + P(2016)$.

2013 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 45

2. 95

3. $\sqrt{65}$ (Must be this exact answer.)

4. -18

5. $\frac{1}{3}$ (Must be this reduced common fraction.)

6. $-6\sqrt{3}$ (Must be this exact answer.)

7. $17\sqrt{19}$ (Must be this exact answer.)

8. -10

9. 1

10. 1

11. $\frac{4}{5}$ (Must be this reduced common fraction.)

12. 24

13. $\frac{297}{625}$ (Must be this reduced common fraction.)

14. 172

15. 16

16. 12870

17. $\frac{56}{33}$ (Must be this reduced improper fraction.)

18. A (Must be this capital letter.)

19. 10

20. 2

ITEM ANALYSIS 3AA PRE-CALCULUS

157 Papers			
% correct			
1. 92%	11. 82%		
2. 97%	12. 25%		
3. 47%	13. 3%		
4. 42%	14. 4%		
5. 89%	15. 89%		
6. 52%	16. 3%		
7. 6%	17. 27%		
8. 43%	18. 93%		
9. 6%	19. 95%		
10. 85%	20. 24%		

4AA PRE-CALCULUS

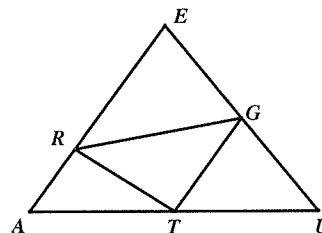
168 Papers			
% correct			
1. 97%	11. 93%		
2. 99%	12. 42%		
3. 69%	13. 8%		
4. 59%	14. 9%		
5. 97%	15. 92%		
6. 66%	16. 14%		
7. 18%	17. 47%		
8. 63%	18. 98%		
9. 14%	19. 97%		
10. 92%	20. 35%		

1. Find the largest real value for x that satisfies $x^{\log_8 x} = \frac{x^3}{64}$.
2. Given $\triangle ABC$ with $AB = 10$, $AC = 15$, and $BC = 21$. Find the product $(\sin B)(\csc A)$.
Express your answer as a common or improper fraction reduced to lowest terms.
3. If $x^5 + 5x^4 + 10x^3 + 10x^2 - 8x + 2 = 14$, find the numerical value of $(x+1)^4$. (Assume $x \neq -1$).
4. Evaluate $\sin^2 15^\circ + \cos^2 75^\circ$. Express your answer as a single reduced rational expression.
5. A sequence is defined recursively as follows: $a_1 = 5$, $a_2 = 9$, $a_n = \frac{2a_{n-1} + a_{n-2}}{3}$. Evaluate $\lim_{n \rightarrow \infty} a_n$.
6. If $2^{x^2+2x} = 7$, then the exact value of $(x+1)^2$ is $\log_2 k$ where k is an integer. Find the value of k .
7. A committee of 3 is selected from 2 juniors and 3 seniors. Find the probability that there are more seniors than juniors on the committee. Express your answer as a common fraction reduced to lowest terms.

8. $i = \sqrt{-1}$. The vector from the origin representing $-8 + 6i\sqrt{3}$ in the complex plane is rotated 150° clockwise about the origin. Find the exact complex number the vector now represents. Express your answer in the form $a + bi$ with exact coefficients a and b .

9. Find the largest integer value of x satisfying $\binom{2013}{1000} + \binom{2013}{1001} = \binom{2014}{x}$.

10. $\triangle AEU$ is equilateral, $\triangle RTG$ has a right angle at T , $RA = 1$, $AT = 4$, and $UT = 2$. Find the exact length of UG . Express your answer as a common or improper fraction reduced to lowest terms.



11. In the expansion of $(x + y)^{12}$ written in decreasing degree for x , the second and third terms are equal when evaluated for $x = a$ and $y = b$, where a and b are positive and $a + b = 1$. Find the value of b . Express your answer as a common fraction reduced to lowest terms.

12. Evaluate $\sum_{n=1}^{1000} \frac{1}{n^2 + 3n + 2}$. Express your answer as a common fraction reduced to lowest terms.

13. For all real numbers x , $f(x^2 + 1) + 3f(22 - 5x) = x^2 + 2$. Find $f(17)$.

14. Find the exact value of $x^2 + 2xy + y^2$ if $\frac{x}{3-5i} + \frac{y}{2+7i} = \frac{5+2i}{41+11i}$, where x and y are both real numbers and $i = \sqrt{-1}$.
15. From the set $\{9, 15, 15, 16, 18, 20, 21, 22, 23, 31, 32\}$, one number is selected at random. Find the probability that the number is within one (population) standard deviation from the arithmetic mean of the set. Express your answer as a common fraction reduced to lowest terms.
16. r and s are the roots of $x^2 + x + 7 = 0$. Find the exact value of $2r^2 + rs + 2s^2 + 7$.
17. Three distinct positive integers form an arithmetic sequence. If the sum of the cubes of the three numbers is divided by the sum of the three numbers, the quotient is 81. If the three numbers are arranged in numerical order, find the second term (the “middle” value) of the sequence.
18. $i = \sqrt{-1}$. A cubic equation with rational coefficients has $1 - 3i$ as one of its roots, 1 as its leading coefficient, and no quadratic term. Find the constant term.
19. If $\left(\sqrt{20\sqrt{14}+5} - \sqrt{20\sqrt{14}-5}\right)^2 = a\sqrt{14} - c\sqrt{d}$, where a , c , and d are positive integers and d is as small as possible, find the exact value of $\sqrt{a+c+d+15}$.
20. Given that $\sin x + \cos x = \frac{6}{5}$ where $0 \leq x \leq \frac{\pi}{6}$, then the exact value of $\sin x$ is $\frac{k+w\sqrt{p}}{q}$ when completely simplified and reduced and where k , w , p , and q are integers. Find the value of $(k+w+p+q)$.

2014 SAA

PreCalculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

- | | |
|---|--|
| 1. <u>64</u> | 11. <u>$\frac{2}{13}$</u> (Must be this reduced common fraction.) |
| 2. <u>$\frac{5}{7}$</u> (Must be this reduced common fraction.) | 12. <u>$\frac{250}{501}$</u> (Must be this reduced common fraction.) |
| 3. <u>13</u> | 13. <u>$-\frac{9}{8}$ OR $-\frac{9}{8}$ OR -1.125</u> |
| 4. <u>$\frac{2-\sqrt{3}}{2}$ OR $\frac{-\sqrt{3}+2}{2}$</u> | 14. <u>4</u> |
| 5. <u>8</u> | 15. <u>$\frac{8}{11}$</u> (Must be this reduced common fraction.) |
| 6. <u>14</u> | 16. <u>-12</u> |
| 7. <u>$\frac{7}{10}$</u> (Must be this reduced common fraction.) | 17. <u>7</u> |
| 8. <u>$7\sqrt{3} - 5i$</u> (Must be this exact answer.) | 18. <u>20</u> |
| 9. <u>1013</u> | 19. <u>$12\sqrt{2}$</u> (Must be this exact answer.) |
| 10. <u>$\frac{14}{5}$</u> (Must be this reduced improper fraction.) | 20. <u>29</u> |

1. Determine the dot product (inner product) between the vectors $\vec{u} = \langle 5, 3 \rangle$ and $\vec{v} = \langle 12, 4 \rangle$.
2. Determine the value(s) for x such that $f(g(x)) = f(x)$ when $f(x) = \ln(x^2 - 2)$ and $g(x) = 2x - 3$.
3. Determine the sum of the value(s) for k such that $(x^4 - k^2x + 3 - k) \div (x - 3)$ leaves a remainder of 4. Express your answer as a common or improper fraction reduced to lowest terms.
4. An ellipse has one vertex at $(6, 0)$, its center at $(0, 0)$, and one focus at $(-4, 0)$. The equation of this ellipse can be expressed in the form $\frac{x^2}{k} + \frac{y^2}{w} = 1$. Determine the value of $(5k + 2w)$.
5. For all t in the set of Real Numbers, $x = 3 - t$ and $y = t^2 - 2t$. Determine the minimum value for y .
6. Determine the exact period of the function $f(x) = \left| 3 \sin \left(\pi x + \frac{1}{3} \right) \right| - 5$.
7. Determine all value(s) for x such that $\log_3(x + 11) - \log_3(x + 3) = 2$

8. Determine the degree measure of the acute angle formed by the intersection of the graphs of the equations $y = \frac{3}{5}x - 7$ and $y = -\frac{1}{2}x + 4$. Express your answer as a decimal rounded to four significant digits.
9. Let k represent a positive integer such that $1 < k < 17$. If $f(x) = x^k + 8x^2 - 32x^4$ and $f(x) = f(-x)$ for all real values of x . Determine the **sum** of all possible distinct values of k .
10. Let the infinite series $S = \frac{4}{3} + \frac{6}{9} + \frac{8}{27} + \frac{10}{81} + \dots$. Determine the exact value of S . Express your answer as a common or improper fraction reduced to lowest terms.
11. The distance between the planes $x + 2y + 3z + 8 = 0$ and $x + 2y + 3z + k = 0$ is 10. Determine all possible exact value(s) for k .
12. When $f(x) = 6x + 17$, $f^{-1}(x) = \frac{kx + w}{p}$ where k , w , and p are relatively prime integers and p is positive. Determine the exact value of $(k + w + p)$.
13. The graph of $y = \frac{3x^2 - 5x - 2}{5x^2 - 14x + 8}$ has a vertical asymptote whose equation is $x = k$ and a horizontal asymptote whose equation is $y = w$. There is also a "hole" (removable discontinuity) in the graph at the point $P(r, s)$. Determine the sum $(k + w + r + s)$. Report your answer as a decimal rounded to the nearest thousandth.
14. A certain restaurant has 8 different items on their value menu and advertises that you can mix and match (including repetitions) any 5 items for \$6.95. (For example, you could order 5 burgers or 3 burgers and 2 shakes, or 5 different single items.) Determine the number of unique orders of 5 items that could be placed off the value menu at this restaurant.

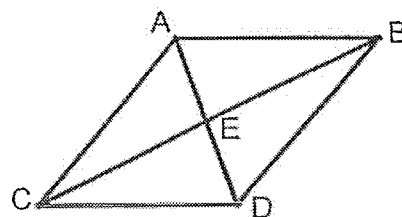
15. Determine the exact value of $\cot \left[\sec^{-1} \left(\frac{-2}{\sqrt{3}} \right) + \csc^{-1}(-2) \right]$.

16. $i = \sqrt{-1}$. $a + bi = (2 - i)^2 + (1 + i)^2$. Determine the exact value of $|a + bi|$.

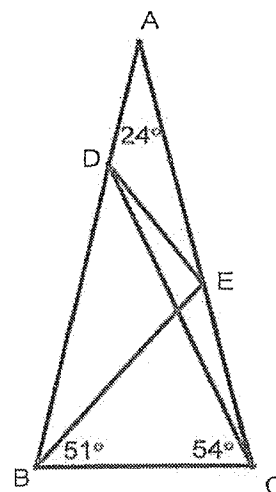
17. Determine the exact value of $\tan(\alpha + \beta + \gamma)$ when $\tan \alpha = \frac{1}{3}$, $\tan \beta = \frac{1}{5}$, and $\tan \gamma = \frac{1}{9}$.
Express your answer as a common or improper fraction reduced to lowest terms.

18. Determine the value of k when $\lim_{n \rightarrow \infty} (\sqrt{n^2 + kn + 5} - \sqrt{n^2 + 3n + 7}) = 4$.

19. Quadrilateral $ABDC$ is a rhombus with sides of length 15. Define $\text{nat} \theta$ to be the length of the diagonal that bisects the angle θ divided by the length of the other diagonal.
 $\text{nat}(\angle ABD) = \frac{4}{3}$. Determine the degree measure of $\angle ACD$. Express your answer rounded to the nearest degree.



20. Triangle ABC is isosceles with $\overline{AB} \cong \overline{AC}$. D lies on $\overline{AB}$ and E lies on $\overline{AC}$. $\angle CAB = 24^\circ$, $\angle EBC = 51^\circ$ and $\angle DCB = 54^\circ$. Determine the degree measure of $\angle EDC$. Report your answer as a decimal rounded to the nearest tenth of a degree.



2015 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

- | | |
|---|---|
| <p>1. <u>72</u></p> <p>2. <u>3</u> (Must be this value only, {3} optional.)</p> <p>3. <u>$-\frac{1}{3}$ OR $-\frac{1}{3}$</u> (Must be this reduced common fraction.)</p> <p>4. <u>220</u></p> <p>5. <u>-1</u></p> <p>6. <u>1</u></p> <p>7. <u>-2</u> (Must be this value only.)</p> <p>8. <u>57.53</u> (Must be this decimal, "degrees" or "°" optional.)</p> <p>9. <u>72</u></p> <p>10. <u>$\frac{5}{2}$</u> (Must be this reduced improper fraction.)</p> | <p>11. <u>$8+10\sqrt{14}, 8-10\sqrt{14}$</u> (Must have both values either order or exact equiv.)</p> <p>12. <u>-10</u></p> <p>13. <u>4.567</u></p> <p>14. <u>792</u> ("possible orders" optional.)</p> <p>15. <u>$-\frac{1}{3}\sqrt{3}$ OR $\frac{-1}{3}\sqrt{3}$ OR $-\frac{\sqrt{3}}{3}$ OR $\frac{-\sqrt{3}}{3}$</u></p> <p>16. <u>$\sqrt{13}$</u> (Must be answer.)</p> <p>17. <u>$\frac{43}{59}$</u> (Must be common)</p> <p>18. <u>11</u></p> <p>19. <u>74</u> ("degrees" or "°" optional.)</p> <p>20. <u>45.3</u> (Must be this exact decimal, "degrees" or "°" optional.)</p> |
|---|---|

ITEM ANALYSIS
3AA PRE-CALCULUS
152 Papers
% correct

1. 79%	11. 4%
2. 57%	12. 74%
3. 34%	13. 52%
4. 34%	14. 3%
5. 85%	15. 44%

6. 46%	16. 40%
7. 93%	17. 41%
8. 48%	18. 36%
9. 74%	19. 36%
10. 62%	20. 1%

4AA PRE-CALCULUS
176 Papers
% correct

1. 87%	11. 25%
2. 66%	12. 82%
3. 45%	13. 62%
4. 57%	14. 16%
5. 88%	15. 56%
6. 52%	16. 70%
7. 92%	17. 59%
8. 60%	18. 53%
9. 81%	19. 54%
10. 69%	20. 5%

1. Determine $\lim_{x \rightarrow \infty} \frac{x^2 - 3x}{3x^2 - 5}$. Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
2. Determine whether the equation $-16x^2 + 9y^2 - 32x - 54y + 209 = 0$ represents a circle, parabola, ellipse, hyperbola, or a degenerate conic. Write the whole word for your answer.
3. Let θ be an acute angle and $\sin 2\theta = k$. Determine the value of $(\sin \theta + \cos \theta)$ in terms of k .
4. Determine the least integral value of n for which the product $\left(100^{\frac{1}{2016}}\right)\left(100^{\frac{2}{2016}}\right)\left(100^{\frac{3}{2016}}\right) \cdots \left(100^{\frac{n}{2016}}\right)$ is greater than 10,000.
5. Determine the sum of the x-coordinate(s) of all ordered pairs of solution(s) for the system of equations
$$\begin{cases} \sqrt{x^2 + y^2} = 1 - y \\ 2y + x = -1 \end{cases}$$
6. The point $(2, 3)$ lies on the graph of $y = f(x - 1)$. Determine the point that must lie on the graph of $y = -f^{-1}(x)$. Express your answer as an ordered pair (x, y) .
7. The graph of the fourth-degree polynomial $P(x)$ is tangent to the x -axis at $(5, 0)$ and $(-5, 0)$. The coefficient of x^4 is 1. The horizontal line $y = k$ meets the graph of $P(x)$ in exactly three points, A , B , and C . Determine the largest of the distances AB , BC , and AC .
8. Let $f(x) = \frac{2x+3}{3x-1}$. Let k represent any value not in the domain of $f(x)$ and let w represent any value not in the range of $f(x)$. Determine the sum $(k + w)$.

9. Given the polar coordinate functions $r_1(\theta) = \frac{4}{\sin \theta + \cos \theta}$ and $r_2(\theta) = \theta$ for $0 \leq \theta \leq 8\pi$.
There are several points common to the graphs of these two functions. Let d be the distance from the common point furthest from the origin to the origin. Then $[d] = n$ where n is a positive integer. Determine the value of n . (Note: $[d]$ is the greatest integer function sometimes called the integer floor function.)
10. Let $g(x) = \cos\left(\frac{1}{2}x\right)$ and $f(x) = \sin 2x$. Determine the maximum value of $(g(x) - f(x))$ over the interval $0 \leq x \leq \pi$. Express your answer as a decimal rounded to four significant digits.
11. Let $\sec x \csc^2 x = 2 \sec x$ where $0 \leq x < 2\pi$. Solution(s) for this equation are in the form $x = k\pi$. Determine the sum of all possible k .
12. The equation $x^3 + 5x^2 + kx + w = 0$ has integer coefficients with $k \neq 0$ and $w = 6k$. One of the roots for this equation is a double root. Determine the sum $(k + w)$.
13. Point $P(8, 0)$ is reflected across the line $y = (\tan(-12^\circ))x$ to get point $Q(a, b)$. Q is then reflected across the line $y = (\tan(18^\circ))x$ to get point $R(c, d)$. The origin is $O(0, 0)$. Determine the exact sum of the lengths of the altitudes of $\triangle POR$.
14. The space curve defined by the parametric equations $x = t$, $y = t^2$, and $z = t^3$, $-\infty < t < \infty$, meets the sphere $x^2 + y^2 + z^2 = 100$ in two points, A and B . With $O = (0, 0, 0)$, determine the degree measure of $\angle AOB$. Express your answer as a decimal rounded to the nearest tenth of a degree.
15. Determine the number of years needed for a sum of money to triple if it is invested at a 5.5% annual rate of interest compounded continuously. Express your answer as a decimal number of years rounded to four significant digits.

16. A circle with radius r has a chord $\overline{AB}$ with positive integer length n . The corresponding minor arc has length $(n+1)$. Determine the smallest possible value of r . Express your answer as a decimal rounded to four significant digits.

17. The Fibonacci sequence is $\{f_n\} = \{1, 1, 2, 3, 5, 8, 13, 21, 34, \dots\}$ where, for $n \geq 2$,

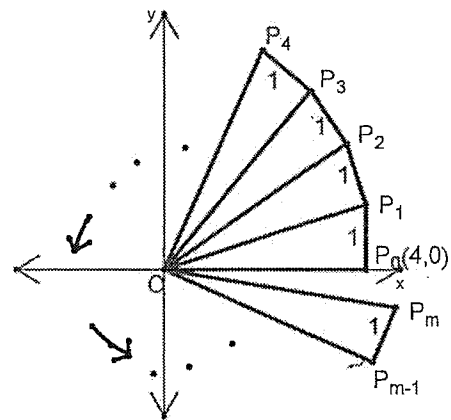
$$f_{n+1} = f_{n-1} + f_n. \text{ The related sequence of quotients is } \{r_n\} = \left\{ \frac{f_{n+1}}{f_n} \right\} = \left\{ \frac{1}{1}, \frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \dots \right\}.$$

Given that $\lim_{n \rightarrow \infty} r_n = L$, determine the absolute difference $|r_{10} - L|$. Express your answer as a decimal rounded to four significant digits.

18. When the expansion of $(3x - 2y)^7$ is completely simplified and written in decreasing degree of x , the numerical coefficient of the 5th term is k . Determine the value of k .

19. In $\triangle ABC$, $\angle CAB = 25^\circ$, $BC = 4$, and $AC = 6$. Determine the least possible length of side $\overline{AB}$. Express your answer as a decimal rounded to four significant digits.

20. In the figure shown, $\angle OP_0P_1$ is a right angle, $OP_0 = 4$ and $P_0P_1 = 1$. Successive right triangles are constructed, with $\angle OP_1P_2 = 90^\circ$, $\angle OP_2P_3 = 90^\circ$, and so on through $\angle OP_{m-1}P_m = 90^\circ$, and all of the indicated segments $P_{m-1}P_m$ have length 1. If P_m is the last point constructed before the next point P_{m+1} lands in the first quadrant, find the length of segment OP_m . Report your answer as a decimal rounded to four significant figures.



2016 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

$\frac{1}{3}$ (Must be this reduced common fraction.)

1. $\frac{1}{3}$

hyperbola (Must be this whole word.)

2. hyperbola

$\sqrt{1+k}$ OR $\sqrt{k+1}$ OR

3. $(k+1)^{\left(\frac{1}{2}\right)}$ OR $(1+k)^{\left(\frac{1}{2}\right)}$

4. 90

5. 1

6. $(3, -1)$ (Must be this ordered pair.)

7. $10\sqrt{2}$ (Must be this exact answer.)

8. 1

9. 24 (Must be this integer.)

10. 1.409 (Must be this exact decimal.)

11. 4

12. -56

13. $12\sqrt{3}$ (Must be this exact answer.)

14. 129.5 ("degrees" or "optional")

15. 19.97 (Must be the "years" or optional.)

16. 1.003 (Must be the exact decimal.)

17. 0.0001478 OR 0.00005646 (Must be one of these decimals.) (units digit of 0 optional)

18. 15,120 (Comma usage optional.)

19. 2.344 (Must be this exact decimal.)

20. 7.141 (Must be this exact decimal.)

ITEM ANALYSIS			
3AA PRE-CALCULUS			
151 Papers			
% correct			
1. 95%	11. 53%		
2. 77%	12. 15%		
3. 16%	13. 1%		
4. 48%	14. 7%		
5. 65%	15. 50%		
6. 37%	16. 0%		
7. 11%	17. 11%		
8. 79%	18. 45%		
9. 18%	19. 30%		
10. 44%	20. 5%		

ITEM ANALYSIS			
4AA PRE-CALCULUS			
144 Papers			
% correct			
1. 95%	11. 59%		
2. 81%	12. 30%		
3. 34%	13. 4%		
4. 68%	14. 20%		
5. 71%	15. 59%		
6. 49%	16. 4%		
7. 42%	17. 24%		
8. 82%	18. 64%		
9. 20%	19. 54%		
10. 57%	20. 9%		

1. (Always, Sometimes, or Never) For your answer, write the whole word "Always", "Sometimes", or "Never"—whichever is correct.

$i = \sqrt{-1}$. Let a , b , c , and d represent real numbers. Then

$$|(a+bi)(c+di)| = (|a+bi|)(|c+di|).$$

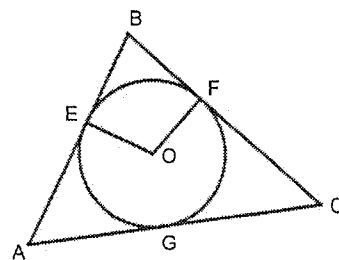
2. Let $k = (\log_{\pi} 8)(\log_{\sqrt{3}} \pi) \left(\frac{1}{2} \log_2 3 \right)$. Determine the exact value of k .

3. Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ represent vectors such that $\vec{a} = (3, 2)$, $\vec{b} = (-8, 13)$ and $\vec{c} = (16, -24)$. Determine the sum of the two vectors that are perpendicular. Express your answer as the ordered pair representation of the sum of those two vectors.

4. Let $f_0(x) = \frac{1}{1-x}$ and $f_n(x) = f_0(f_{(n-1)}(x))$ for integers $n \geq 1$. Determine the value of $f_{2017}(2017)$. Express your answer as an integer or as a common or improper fraction reduced to lowest terms.

5. Determine the sum of the series $\sum_{n=1}^{\infty} \frac{1}{3^n}$, if it exists. Report your answer as an integer or as a common or improper fraction reduced to lowest terms, or "DNE" if the sum does not exist.

6. Circle O is inscribed in $\triangle ABC$ with E , F , and G the respective points of tangency. $AB = 8$, $BC = 10$, and $CA = 12$. Determine the radian measure of $\angle EOF$. Express your answer as a decimal rounded to the nearest thousandth.



7. Determine the remainder when 2^{2017} is divided by 1023.
8. $\text{Arc tan}(x)$ is expressed in radians and represents the inverse tangent function. Determine the smallest value in the range of $f(x) = 8(\text{Arc tan}(x^2 + 1))$.

9. At a party of married couples where every man is married to a woman, every woman is married to a man, and all are present, every man shakes hands with every other man and that man's wife. However, none of the wives shake each other's hands and no man shakes his own wife's hand. 459 handshakes occur. Determine how many people are at the party.

10. $f(x) = \begin{cases} ax^2 + bx + 1 & , \quad x < 3 \\ 4 & , \quad x = 3 \\ ax - \frac{b}{2} & , \quad x > 3 \end{cases}$. Determine the ordered pair (a, b) such that $f(x)$ is continuous at $x = 3$.

11. Let θ be an acute angle such that $\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{x-1}{2x}}$. Then $\tan^2 \theta = kx^2 + wx + p$. Determine the sum $(k + w + p)$.

12. $k = \lim_{x \rightarrow 0} \left(\frac{x + \tan x}{x} \right)$. Determine the value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms..

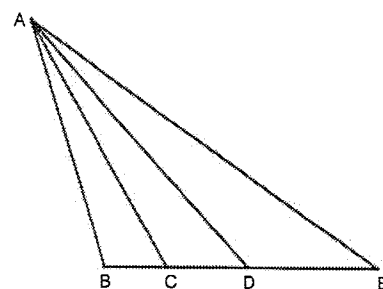
13. $i = \sqrt{-1}$. $f(x) = x^2 + kx + w$ has real coefficients k and w . The zeros of $f(x)$ are r_1 and r_2 such that $r_1 - r_2 = 2i$ and $\frac{1}{r_1} + \frac{1}{r_2} = \frac{3}{5}$. Determine the sum of all possible value(s) for k and w . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.

14. A circular carnival Ferris wheel has a diameter of 20 meters with the lowest point at a ramp 5 meters off level ground. The Ferris wheel makes a complete revolution in 30 seconds. At time $t = 0$, a point on the outside of the Ferris wheel is at its lowest height above ground. Determine, in meters, the height above the ground of that point on the wheel at 17 seconds. Express your answer as a decimal rounded to the nearest thousandth.

15. $5\sin(4x) + 2\cos(4x) = a\sin(bx + c)$. Determine the exact value of the sum $(a + b)$.

16. Let n represent a positive integer such that $\frac{720}{n!} = 6$. Determine the value of n .

17. C and D lie on $\overline{BE}$ such that $\overline{AC}$ and $\overline{AD}$ trisect $\angle BAE$ in $\triangle ABE$. $BC = 2$, $CD = 3$, and $DE = 6$. Then the perimeter P of $\triangle ABC$ may be expressed as $P = f + k\sqrt{w} + p\sqrt{q}$ in simplified and reduced radical form. Determine the sum $(f + k + w + p + q)$. Express your answer as an integer or as a common or improper fraction reduced to lowest terms.



18. Determine the length of the polar curve $r = \frac{4}{\sin \theta + \cos \theta}$ when $0 \leq \theta \leq \frac{\pi}{2}$.

19. The equations of two planes in 3-space are given by $2x - 4y + 2z = -8$ and $x + y - z = 0$. A parametric representation of the line of intersection of these two planes is $x(t) = -2 + t$, $y(t) = at$, and $z(t) = b + ct$ for all real values of t . Determine the sum $(a + b + c)$.

20. Your Math Team Coach probably often says, "There is no i in 'team'." Since Coach is also a math teacher, Coach might say, "There is no $e^{ki\pi}$ in 'team'." (To a math teacher, $i = \sqrt{-1}$.) Given Euler's formula: $e^{i\theta} = \cos \theta + i \sin \theta$, determine the smallest positive value of $(12k - 100)$.

2017 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X **2** pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specification stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

(Must be this whole word, capital A optional.)

1. Always

11. 0 or zero

2. 3

12. 2

3. (19, -22) (Must be this ordered pair.)

13. $\frac{40}{9}$ (Must be this reduced improper fraction.)

4. $\frac{2016}{2017}$ (Must be this reduced common fraction.)

14. 24.135 (Must be this decimal, "meters" or "m" optional.)

5. $\frac{1}{2}$ (Must be this reduced common fraction.)

15. $4 + \sqrt{29}$
OR $\sqrt{29} + 4$ (Must be this exact answer.)

6. 1.696 (Must be this decimal, "radians" or "r" optional.)

16. 5

7. 128 OR 2^7

17. 23

8. 2π (Must be this exact answer.)

18. $4\sqrt{2}$ (Must be this exact answer.)

9. 36 ("people" optional.)

19. 3

10. (1, -2) (Must be this ordered pair.)

20. 2

ITEM ANALYSIS 3AA PRE-CALCULUS

183 Papers	
% correct	
1. 44%	11. 19%
2. 90%	12. 84%
3. 76%	13. 13%
4. 18%	14. 37%
5. 79%	15. 8%
6. 15%	16. 91%
7. 25%	17. 4%
8. 62%	18. 31%
9. 16%	19. 14%
10. 67%	20. 15%

ITEM ANALYSIS 4AA PRE-CALCULUS

176 Papers	
% correct	
1. 55%	11. 27%
2. 92%	12. 88%
3. 86%	13. 26%
4. 40%	14. 36%
5. 91%	15. 22%
6. 33%	16. 96%
7. 48%	17. 5%
8. 74%	18. 41%
9. 36%	19. 30%
10. 77%	20. 19%

1. Let $\begin{vmatrix} 5 & k & 7 \\ 4 & 3 & 0 \\ 0 & 3 & 5 \end{vmatrix} = 39$. Determine the value of k .
2. The point $(12, 7)$ is on the graph of $y = f(x)$. Determine the coordinates of the point that must be on the graph of $y = 4f(x-8) - 10$. Express your answer as an ordered pair (x, y) .
3. Let $k = \lim_{x \rightarrow 4} \frac{2 - \sqrt{x}}{x - 4}$. Determine the value of k , if it exists. Express your answer as an integer or as a common or improper fraction if the limit exists or write "DNE" if it does not exist.
4. Let O be the center of a (concave) 5-point star $APBQCRDSET$ having 5 lines of symmetry. The points A, B, C, D , and E lie on the circumscribed circle with radius 10. The points P, Q, R, S , and T are the corners of the star that are closest to the center O . The points A, P, Q , and C are collinear and $\overline{AR}$ is one of the symmetry lines. Let k be the numerical area of the star and w be the perimeter. Determine the sum $(k + w)$. Round your answer to the nearest integer.
5. Let f be the function defined by $f(x-2) = 4x^3 - 2x^2 + 7x + 13$. Determine the value of the sum $f(5) + f(-4)$.
6. A not-so-gifted trigonometry student "solved" the equation $\cos x = 2x$ with x in radians by "canceling the x 's" and reported $\cos = 2$, causing the teacher to faint. Determine what the student *should* have reported for the value of x . Express your answer as a decimal rounded to four significant digits.
7. The cosine function can be expressed as an infinite series $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{(2n)}}{(2n)!}$ for all $x \in \mathbb{R}$. Determine the least number of terms needed to approximate $\cos(0.5)$ within 0.0001 of the given value for $\cos(0.5)$ from your calculator.

8. $\mathfrak{U} = \tan^2 \theta + \frac{\cos^2 \theta - \cos^4 \theta}{\sin^2 \theta \cos^2 \theta}$ where $\mathfrak{U}$ is equal to an expression having only one term using only one of the six ratios $\sin \theta$, $\cos \theta$, $\tan \theta$, $\cot \theta$, $\sec \theta$, or $\csc \theta$ (and the denominator is one.) Assume non-zero denominators. Determine $\mathfrak{U}$ and write this full expression as your answer. (Do not include the denominator of one.)
9. A group of five men and five women are seated randomly at the ten chairs around a circular table. Determine the probability that at least two men are seated next to each other or that at least two women are seated next to each other. Express your answer as an integer or as a common fraction.
10. The graph of the polar equation $r = 2(1 - \sin \theta)$ on a standard grid is a cardioid (heart-shaped). A circle is circumscribed about this cardioid so that the two figures are tangent at the right and left extremes of the cardioid. Determine the exact radius of the circle.
11. $f(x) = \frac{x-2}{x+2}$ and $g(x) = f(f(f(x)))$. The domain for $g(x)$ is all real numbers except for some exclusion(s). Determine the sum of the excluded value(s) from the domain of $g(x)$. Express your answer as an integer or as a common or improper fraction.
12. Lines L_1 and L_2 are given parametrically by the equations $x = 5 + t$, $y = 6 - 2t$, $z = -10 + 3t$ and $x = 2t - 2$, $y = 5 - t$, $z = 9 - 2t$, respectively. Determine the point of intersection of these two lines. Report the ordered triple (x, y, z) for the common point, or if they do not intersect, report "NONE".
13. The graphs of $x^5 + y^5 = 1$ and $x + y = k$ have exactly one point of intersection in Quadrant I. The exact x-coordinate of this point can be written as $p^{(w)}$ where p is a positive prime integer and w is a rational number. Determine the sum $(p + w)$. Express your answer as an integer or as a common or improper fraction.
14. Given $P(x) = ax^2 + bx + c$ where the sum of the x -intercepts is 3.5, the product of the x -intercepts is -2.5 and $P(-1) = 10$. Determine the value of $P(2)$. Express your answer as an integer or as a common or improper fraction.

15. The general equation of the plane containing the three points $(2,1,3)$, $(-3,7,-1)$, and $(4,-2,5)$ can be written as $Ax + By + Cz = D$ where A , B , C , and D are relatively prime integers with $D > 0$. Determine the sum $(A + B + C + D)$.
16. A curve is given parametrically by the equations $x = 10 \cos t$, $y = 10 \sin t$, and the restriction $0 \leq t \leq \frac{2}{3}\pi$. The exact length of this curve is $k\pi$. Determine the exact value of k . Express your answer as an integer or as a common or improper fraction.
17. Suppose that the statement " $\sin(x) = (\text{six})n$ for some integer n and x in radians" makes trigonometric sense. Determine the sum of the 20 smallest non-negative values of x . (Note: " $(\text{six})n$ " means "six times n ".)
18. Assume that on any single throw of a dart at the dartboard, Bob will hit the bulls-eye 75% of the time. Assume that on any single throw of a dart at the dartboard, Jack will hit the bulls-eye 60% of the time. Bob and Jack agree to alternate throws until one hits the bulls-eye. If Bob goes first, determine the probability that Bob will hit the bulls-eye before Jack does. Express your answer as an integer or as a common fraction.
19. The degree measure of one angle of a parallelogram is four times the degree measure of a second angle. The side lengths in inches are $\sin 90^\circ$ and $-4 \cos 180^\circ$. Determine the numerical area of the parallelogram in square inches. Express your answer as a decimal rounded to the nearest thousandth.
20. A geometric sequence has three non-zero terms. The middle term is $2 \sin \theta$, the common ratio is $2 \cos \theta$, and the last term is twice the first term. Values for θ , $0 \leq \theta \leq 2\pi$, that fit this sequence are of the form $k\pi$. Determine the largest value for k . Express your answer as an integer or as a common or improper fraction.

2018 SAA

Pre-Calculus

Name ANSWERS

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. 6

11. $-\frac{34}{15}$ OR $-\frac{34}{15}$ (Must be this reduced improper fraction.)

2. (20,18) (Must be this ordered pair.)

12. (8,0,-1) (Must be this ordered triple.)

3. $-\frac{1}{4}$ OR $-\frac{1}{4}$ (Must be this reduced common fraction.)

13. $\frac{9}{5}$ (Must be this reduced improper fraction.)

4. 185

14. $-\frac{55}{2}$ OR $-\frac{55}{2}$ (Must be this reduced improper fraction.)

5. 1295

15. 16

6. 0.4502
OR .4502 (Must be this decimal.)

16. $\frac{20}{3}$ (Must be this reduced improper fraction.)

7. 3 (“terms” optional.)

17. 190π

8. $\sec^2 \theta$ OR $(\sec \theta)^2$ (Must be this expression.)

18. $\frac{5}{6}$ (Must be this reduced common fraction.)

9. $\frac{125}{126}$ (Must be this reduced common fraction.)

19. 2.351 (Must be this decimal, “square inches” or “sq. in.” optional.)

10. $\frac{1}{2}(3\sqrt{3})$ OR $\frac{3}{2}\sqrt{3}$ OR $\frac{3\sqrt{3}}{2}$

20. $\frac{7}{4}$ (Must be this reduced improper fraction.)

1. The lengths of the sides of a right triangle are all whole numbers. Two of the lengths are 48 and 55. Determine the value of the cosine of the smallest angle of this right triangle. Express your answer as an integer or as a common or improper fraction.
2. The function f is defined as $f(x) = \frac{x^2 + x - 380}{x^2 - 35x + 304}$. Then $\lim_{x \rightarrow 19} f(x) = k$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
3. $\log_8 \begin{vmatrix} 1 & 3 & -1 \\ 2 & 0 & 2 \\ -3 & -4 & -3 \end{vmatrix} = k$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
4. The function f is defined as $f(x) = \frac{3x}{2x+5}$. Then $f^{-1}(x) = \frac{kx+w}{px+q}$ in simplified rational form with integers k , w , p , and q with $p > 0$. Determine the sum $(k + w + p + q)$.
5. A regular pentagon has area and perimeter that are numerically equal. Determine the length of one side of the pentagon. Express your answer as a decimal rounded to the nearest thousandth.
6. The vectors $\vec{v} = -4\vec{i} - 8\vec{j}$ and $\vec{w} = 44\vec{i} + k\vec{j}$ are perpendicular. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
7. For $k > 4$, there is a region in the first quadrant bordered by the line $x + y = k$ and the hyperbola $xy = k$. The graph of $y = |x|$ meets the line and the hyperbola in points P and Q . Determine the value of k that makes $PQ = 1$. Express your answer as a decimal rounded to the nearest thousandth.

8. The line $3x + 4y + 7 = 0$ makes an angle of inclination of θ degrees with the positive x-axis. Determine the value of θ . Express your answer as a decimal rounded to the nearest thousandth.
9. $i = \sqrt{-1}$. Then, the complex number $z = 64 \left(\cos \left(\frac{5\pi}{6} \right) + i \sin \left(\frac{5\pi}{6} \right) \right)$ has $\sqrt{z} = a + bi$ for some real numbers a and b . Determine the sum $(a+b)$. Express your answer as a decimal rounded to the nearest thousandth.
10. A triangle has side-lengths of 5, 7, and 6. Determine the value of the cosine of the smallest angle of this triangle. Express your answer as an integer or as a common or improper fraction.
11. k_1 is the smallest positive integer and k_2 the largest negative integer for which the equation $x^3 + kx^2 + 6x = 0$ has more than one real non-zero solution. Determine the product $(k_1 k_2)$.
12. The period of the graph of $y = 20 \tan(19x - 38) + 12$ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
13. A five-pointed star is formed from the diagonals of a regular pentagon with edge length 12. Let k be the area of the star. Determine the value of k . Express your answer as a decimal rounded to the nearest thousandth.
14. Determine the exact sum $\sum_{n=10}^{\infty} \left(\frac{4}{n^2 - n - 6} \right)$. Express your answer as an integer or as a common or improper fraction.

15. The exact sum of all solutions to $\sin^2 x = \cos^2 x$ when $16\pi \leq x \leq 17\pi$ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
16. Two planes, $3x - 2y + 5z = 20$ and $x + 2y + 3z = 12$, intersect in a line. The parametric equations for this line can be expressed as
$$\begin{cases} x = a + kt \\ y = b + wt \\ z = -2t \end{cases}$$
 for all real t . Determine the value of the product $(a+b)(k+w)$.
17. A three-term sequence $\{a_n\}$ is defined by the relation: a_n is the solution for x to the equation $e^{\log x} = e^n$, $n = 1, 2, 3$. A three-term sequence $\{b_n\}$ is defined by the relation: b_n is the solution for y to the equation $10^{\ln y} = 10^{4-n}$, $n = 1, 2, 3$. Let the points P_n be (a_n, b_n) , $n = 1, 2, 3$. The exact area of triangle $P_1P_2P_3$ is a polynomial in e . Determine the exact sum of the numerical coefficients in this polynomial.
18. The double-helix model of a DNA molecule is mathematically quite complicated. A simplified version is represented by the two sets of parametric equations $(x = 3 \cos t, y = 3 \sin t, z = 5t)$ and $(x = 3 \cos(t + \pi), y = 3 \sin(t + \pi), z = 5t)$, $0 \leq t \leq 6\pi$. For each value of t , the segment joining the corresponding points on the two helices has a constant length L . Determine the value of L .
19. The area between the x -axis and the graph of $y = |\sin x|$ between $x = 0$ and $x = k$, for $0 \leq k \leq \pi$, is given by $|1 - \cos k|$. The area between the x -axis and the graph of $y = |\sin x|$ between $x = \frac{\pi}{2}$ and $x = w$ is 5.4. Determine the value of w . Express your answer as a decimal rounded to the nearest thousandth.
20. Let k represent a positive integer such that $3 < k < 1023$. Determine the largest possible value of k such that $\sin^2(k^\circ) + \cos^2(k^\circ) - 1 = 0$.

2019 SAA

Name ANSWERS

Pre-Calculus

School _____

(Use full school name – no abbreviations)

_____ Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. $\frac{55}{73}$ (Must be this reduced common fraction.)

11. -25

2. 13

12. $\frac{1}{19}$ (Must be this reduced common fraction.)

3. $\frac{4}{3}$ (Must be this reduced improper fraction.)

13. 116.971 (Must be this decimal.)

4. -6

14. $\frac{15797}{34650}$ (Must be this reduced common fraction.)

5. 2.906 (Must be this decimal.)

15. 33

6. -22

16. 50

7. 6.522 (Must be this decimal.)
(Must be this decimal, "degrees" or "°" optional. Trailing zero necessary.)

17. 0 OR "zero"

8. 143.130

18. 6

9. 9.798 (Must be this decimal.)

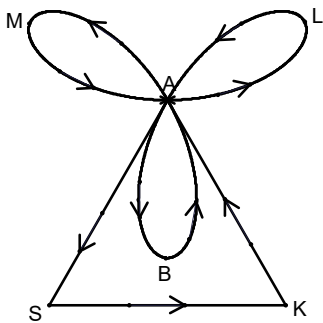
19. 10.352 (Must be this decimal.)

10. $\frac{5}{7}$ (Must be this reduced common fraction.)

20. 1022

1. Let $\sin x = 0.6$ and $\tan x > 0.3$. Determine the value of $\cos x$. Express your answer as an exact decimal.
2. Let $k = \lim_{x \rightarrow 10} x\sqrt{x+2}$. Determine the value of k .
3. Let $M = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ and n is a positive integer. Determine the sum of the elements of M^n .
Express your answer as a simplified exponential expression in terms of n .
4. For x an integer and $x > 4$, the expression $\frac{x!(x-3)!}{(x-5)!(x+2)!}$ can be rewritten as the simplified rational expression $\frac{ax^2 + bx + c}{mx^2 + nx + p}$ with integer coefficients. Determine the value of $\left(\frac{b}{n}\right)$.
Express your answer as an integer or as a common or improper fraction.
5. Determine the coordinates of the point $A(x, y, z)$ that is two-fifths of the way from point $P(5, -1, 4)$ to point $Q(2, 3, 1)$. Express your answer as that ordered triple (x, y, z) using an integer or a common or improper fraction for the coordinate entries.
6. The numbers a , b , c , and d can each be used once to fill the entries of a 2×2 matrix. Determine the maximum number of distinct determinant values that could be created.
7. Let f be the function given by $f(x) = 2x + 1$, g be the function given by $g(x) = x^2 + 2x - 17$, and g^{-1} the inverse relation for the function g . Determine the sum of all x such that $f(x) = g^{-1}(x)$. Express your answer as an integer or as a common or improper fraction.
8. Determine the number of distinct quintuples of positive integers (a, b, c, d, e) which exist when $a + b + c + d + e = 16$.

9. A large school has 2022 parking spaces each tagged sequentially from 1 to 2022. As a senior prank, a group of students removed the tags whose numbers were divisible by 2, 3, or 7. Determine the number of tags that were removed.
10. The parametric equations $x = e^t \cos t$ and $y = e^t \sin t$, with t in radians and $t \geq 0$, define a spiral curve. The slope m of the tangent line to this curve at the point (x, y) is given by the formula $m(t) = \frac{\cos t + \sin t}{\cos t - \sin t}$. As t increases from 0, let P be the first point at which the tangent line is vertical and let Q be the first point at which the tangent line is horizontal. Determine the distance PQ . Express your answer as a decimal rounded to the nearest thousandth.
11. David has three equally-sized rectangular sheets of paper: one blue, one white, and one red. He cuts out a white ellipse and glues it on the blue sheet so that it just touches the midpoint of, and is tangent to, all four sides of the blue rectangle. He then cuts the red piece into three congruent squares and glues them, without overlap, on top of his design, completely covering both the blue rectangle and the white ellipse. David finds that the length of the diagonal of one of the red squares is $10\sqrt{2}$ inches. Determine the number of square inches of the scrap white paper that remains. Express your answer as a decimal rounded to the nearest thousandth.
12. In $\triangle ABC$, $AB = 7$, $AC = 3$, $CB = 6$, and $\overline{AD}$ is a median of the triangle. Determine the length of $\overline{AD}$.
13. The planes with equations $4x - 3y + 12z = 89$ and $-3x - 12y + 4z = 10$ meet to form an acute dihedral angle of θ degrees. Determine the degree value of θ . Express your answer as a decimal rounded to the nearest thousandth.
14. There are many arrangements of the ten letters in the word EXPERIMENT. Some of those ten-letter arrangements have the math word PRIME appearing in that order, with no other letters mixed in between. Determine the number of arrangements that satisfy this condition.
15. A pilot's eyes are at a vertical height of k feet above a horizontal straight-line path to a 6000-foot runway. At one instant, the angle of depression of the pilot's eyes to the nearer end of the runway is 33° and the angle of depression from the pilot's eyes to the far end of the runway is 15° . Determine the vertical height k , in feet, of the pilot's eyes at that instant. Express your answer rounded to the nearest integer.

16. Determine the value of x when $2 \operatorname{Arctan}\left(\frac{1}{5}\right) + \operatorname{Arctan} x = \frac{\pi}{4}$. Express your answer as an integer or as a common or improper fraction.
17. The roots of $3y^3 + 2y^2 + y - 1 = 0$ are a , b , and c . Then $kx^3 + wx^2 + px + q = 0$, for relatively prime integers k , w , p and q with $k > 0$, is a polynomial in terms of x whose roots are $a+2$, $b+2$ and $c+2$. Determine the sum $(k+w+p+q)$.
18. Thirteen unit circles are arranged with their centers equally spaced on a circle of radius r in such a way that each of the unit circles is externally tangent to exactly two others. Determine the value of r . Express your answer as a decimal rounded to the nearest thousandth.
19. In radian mode the polar coordinate equation $r = 2 \sin(3\theta)$, $0 \leq \theta \leq \pi$, produces the three-leaved rose designated as *ALABAMA*. The “curve” is traced out as it starts at A and passes through L , A , B , A , M and ends back at A in that order. Points L , B , and M are vertices of an equilateral triangle. The total area inside the rose is given to be π . A second “curve” consisting of a leaf of the rose and a triangle designated as *ALASKA*, is similarly traced. It starts at A and traces the upper right leaf of the rose through L , returning to A , and then passing down tangent (to the rose) segment $\overline{AS}$, through points S and K such that $AS = AK$, and ending back at A along tangent segment $\overline{AK}$. (Diagram not drawn to scale.) When the total areas enclosed by the *ALABAMA* and *ALASKA* “curves” are equal, the distance $SK = 2^k 3^w \pi^p$. Determine the ordered triple (k, w, p) . Express your answer as that ordered triple (k, w, p) using an integer or a common or improper fraction for the coordinate entries.
- 
20. A particle follows a path mapped by the parametric equations $x(t) = 20(t-1)$ and $y(t) = -t^2 + 22t + 23$, where $x(t)$ is the horizontal position and $y(t)$ is the vertical position. The particle starts at time $t = 0$ and moves to the point where it reaches maximum height. Determine the number of units this particle moved horizontally. Express your answer as an integer or as a common or improper fraction.

Pre-Calculus

School

(Use full school name – no abbreviations)

Correct X 2 pts. ea. =

Note: All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1. $(\cos x =) 0.8$ OR $.8$ (This exact decimal.) 11. 64.381 (This decimal.) (square inches)

2. $(k =) 20\sqrt{3}$ (This exact answer.) (limit) 12. $(AD =) 2\sqrt{5}$ (This exact answer.) (median length)

3. $2^{(n+1)}$ OR $2^{(1+n)}$ (Must be this expression, parenthesis around exponent optional.) 13. 64.784 (This decimal.) (degrees or "°")

4. $(b/n =) -\frac{7}{3}$ OR $-\frac{7}{3}$ (This fraction.) (ratio) 14. 360 (This integer.) (anagrams)

5. $(A(x,y,z) =) \left(\frac{19}{5}, \frac{3}{5}, \frac{14}{5}\right)$ (This ordered triple with these fractions as entries.) 15. $(k =) 2737$ (This integer.) (height in feet)

6. 6 (This integer.) (determinant values) 16. $(x =) \frac{7}{17}$ (This fraction.)

7. $-\frac{7}{4}$ OR $-\frac{7}{4}$ (This fraction.) (Note: $x =$ is incorrect.) 17. -3 (This integer.)

8. 1365 (This integer.) (Comma use ok.) (quintuples) 18. $(r =) 4.179$ (This decimal.) (radius)

9. 1444 (This integer.) (Comma use ok.) (tags removed) 19. $((k,w,p) =) \left(\frac{3}{2}, -\frac{3}{4}, \frac{1}{2}\right)$ OR $\left(\frac{3}{2}, \frac{-3}{4}, \frac{1}{2}\right)$ (Must be this ordered triple with these fractions.)

10. $(PQ =) 10.776$ (This decimal.) (distance) 20. 220 (This integer.) (units) ($x =$ or $x(t) =$ not correct.)